
IDENTITY IS THE SIGNAL: WHY PRE-EXECUTION AGENT-ACTION RISK NEEDS DELEGATED AUTHORITY, NOT JUST CONTENT

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ABSTRACT

Guardrail models for AI agents typically classify a proposed action from its text alone. We show this misses the dimensions that matter most for delegated, tool-using agents. Stripping principal entitlements, delegated scopes, and policy text from a trained 2B-parameter decoder’s input drops macro AUPRC from 0.915 to 0.736 on held-out tools, concentrated entirely in three dimensions — unauthorized-scope detection falls from 1.00 to 0.12, policy-conflict detection from 1.00 to 0.35 — while two zero-shot safety models (Granite Guardian 3.3 8B, gpt-oss-safeguard 20B) sit at chance on exactly these dimensions with or without that context, showing the gap is not something prompting a general safety model can close. We further show that the *shape* of policy-conditioned reasoning, not just tool identity, must generalize to unseen policy kinds and phrasings, and that naive multi-policy composition collapses to a 0.97 false positive rate by three stacked policies unless decisions are made from calibrated joint probabilities rather than independent thresholds. All results are on synthetic data; we are explicit about what that does and does not license.

1 INTRODUCTION

Deployed AI agents increasingly act with delegated authority: a support agent issues refunds for a merchant, a coding agent pushes commits for a developer, an operations agent grants IAM roles for an administrator. Before such an action executes, some system must decide whether it is safe to proceed. The dominant tool is a guardrail classifier that reads the proposed action and outputs a harm judgment (Inan et al., 2023; Padhi et al., 2024; Zeng et al., 2024; Ghosh et al., 2024; OpenAI, 2025).

We argue this conflates two questions. **Content risk** asks whether an action looks dangerous on its face (deletes data, moves money, contacts an external party) and is answerable from the action text alone. **Authorization risk** asks whether *this* action is something *this* agent was allowed to do, on behalf of *this* principal, under the policy in force – and requires the principal’s entitlements, the agent’s delegated scope, and the applicable policy clause as structured inputs, not prose the model must infer.

We build a pre-execution risk classifier, *forecheck*, whose input contract makes these fields explicit, then strip them from the same trained model’s input to see what breaks. On synthetic data, three of eleven risk dimensions collapse from near-perfect to near-chance when identity is removed, while dimensions already recoverable from action text alone are unaffected. This is not a fine-tuning artifact prompting could fix: two zero-shot safety models given the same policy text in-prompt are at chance on exactly these dimensions, with or without identity, because they were never trained to use it. We further test whether policy-conditioned reasoning generalizes to unseen policy *kinds* and *phrasings*, and show that composing several such classifiers’ decisions – how guardrail stacks are deployed in practice – fails in a specific, fixable way: per-policy thresholding compounds toward denying everything as more policies are added, while deciding from the joint calibrated probability vector does not.

Contributions. (1) An identity-ablation diagnostic separating which of eleven risk dimensions need delegated-authority context from which are content lookups (§3); (2) evidence that two current zero-

shot safety models do not use delegated-authority context even when it is handed to them (§3); (3) a policy-generalization result showing robustness to unseen policy wording but not unseen policy kinds (§4); (4) a multi-policy composition result showing calibrated joint-probability decisions, not rule engineering, keep a guardrail stack from collapsing to "deny everything" (§5); and (5) a comparison showing an agent asked to judge its own action cannot recover the authorization signal an external checker can (§6). Every number here is on synthetic data from forecheck’s own scenario generator; §8 states plainly what that does and does not license.

2 PROBLEM SETTING

Contract. forecheck scores a proposed tool call against an `ActionContext`: the principal’s entitlements, the scope actually delegated to the agent (a possibly strict subset of the principal’s), a trust-labelled trajectory of prior observations, the applicable policy clause(s), and the proposed action (tool, arguments, destination) – typed fields the classifier reads directly, not facts inferred from prose. A classifier that sees only the rendered action text, as every guardrail model in §7 does, can answer content-risk questions but has no way to check whether the agent was delegated the scope it is exercising.

Eleven risk dimensions. Four are *content/consequence* dimensions decidable from the action and destination alone: `destructive_or_irreversible_action`, `financial_commitment`, `external_communication`, `untrusted_destination`. Two are *trajectory* dimensions decidable from prior observations without identity: `prompt_injection_influence`, `suspicious_action_sequence`. The remaining are where identity and policy are load-bearing: `unauthorized_scope` (inside the principal’s entitlements *and* the agent’s delegated scope?), `policy_conflict` (violates a stated clause?), `insufficient_context` (enough delegated information to decide at all?), and `privilege_escalation` (does the action widen authority beyond what was granted?). `sensitive_data_exposure` behaves like a content dimension in practice (§3) because its deciding field renders directly into the action arguments.

Data. All results are on synthetic data from forecheck’s own scenario generator (49,996 rows at the v2/v3/v4 scale used here), split into `train/calibration/ dev/test`, a `heldout_family` split withholding a stratified subset of tools, and, from v3 on, `heldout_policy_kind/heldout_policy_phrasing` splits (§4). Every number measures whether an arm learned the generator’s label function, not real-world risk; no human-labelled or production-traffic evaluation exists at submission time (§8).

Arms. A LoRA-adapted 2B decoder (`granite-3.3-2b-instruct`) scored by candidate-token logits over the eleven dimensions (Hu et al., 2021); two linear-head encoders (ModernBERT-large (Warner et al., 2024), `granite-embedding-r2`); a deterministic rule baseline with no learned component; two zero-shot safety models with the same policy text in-prompt and no fine-tuning, Granite Guardian 3.3 8B (Padhi et al., 2024) and `gpt-oss-safeguard` 20B (OpenAI, 2025); and an *agent self-judgment* condition where an untrained instruct model (2B, 8B (IBM Granite Team, 2024)) gets the same context in a Fin-style system prompt and returns one ALLOW/STOP verdict (§6).

3 IDENTITY ABLATION

At evaluation time only, we remove principal entitlements, delegated scope, `on_behalf_of`, and policy text from the trained decoder’s rendered context (`-strip-identity`), leaving action, destination, and trajectory untouched, and compare against a rule baseline (action and destination fields only) and two zero-shot safety models given the same policy text in-prompt.

Table 1: identity-aware trained models lose 0.10–0.18 macro AUPRC when identity is removed; the rule baseline is unchanged; the zero-shot models are flat or slightly *improve*. If a general safety model quietly used delegated-authority context, stripping it would hurt – it does not, for either model.

Figure 1 shows the same pattern across all eleven dimensions: the drop from stripping identity is concentrated in `unauthorized_scope`, `policy_conflict`, and `insufficient_context`, while the remaining dimensions barely move.

Table 1: Macro AUPRC with and without identity/policy context. Decoder and encoder rows are on `heldout_family` unless noted; zero-shot rows use a 2,000-row (Granite Guardian) or 400-row (gpt-oss-safeguard) subsample. Zero-shot models are unaffected or slightly helped by stripping, showing they never used the context. Source: `docs/results/synthetic-v2/README.md`.

Arm	Split	Macro AUPRC (full)	Macro AUPRC (stripped)
Decoder 2B v2	heldout_family	0.915	0.736
Decoder 2B v4 ^a	heldout_family	0.960	—
Encoder ModernBERT-large	heldout_family	0.813	0.712
Encoder granite-embedding-r2	heldout_family	0.841	—
Granite Guardian 3.3 8B (zero-shot)	heldout_family	0.257	0.271
gpt-oss-safeguard 20B (zero-shot)	test (n=400)	0.269	0.238
Agent self-judgment 8B ^b	heldout_family (n=2000)	0.242	0.281
Rule baseline	heldout_family	0.614	0.614

^aIdentity-stripped v4 run [TODO: pending: decoder-2b-v4 -strip-identity job]. ^bSingle ALLOW/STOP verdict scored against all eleven labels (§6), not a per-dimension classifier.

Table 2: Decoder 2B v2, `heldout_family` AUPRC per dimension: full, identity-stripped, and rule baseline. First three rows are delegated-authority dimensions; the last is a representative content/lookup dimension. Source: `docs/results/synthetic-v2/README.md`.

Dimension	Decoder (full)	Decoder (stripped)	Rule baseline
<code>unauthorized_scope</code>	1.000	0.116	0.512
<code>policy_conflict</code>	0.999	0.345	0.315
<code>insufficient_context</code>	0.849	0.418	0.235
<code>financial_commitment</code> (lookup)	1.000	1.000	1.000

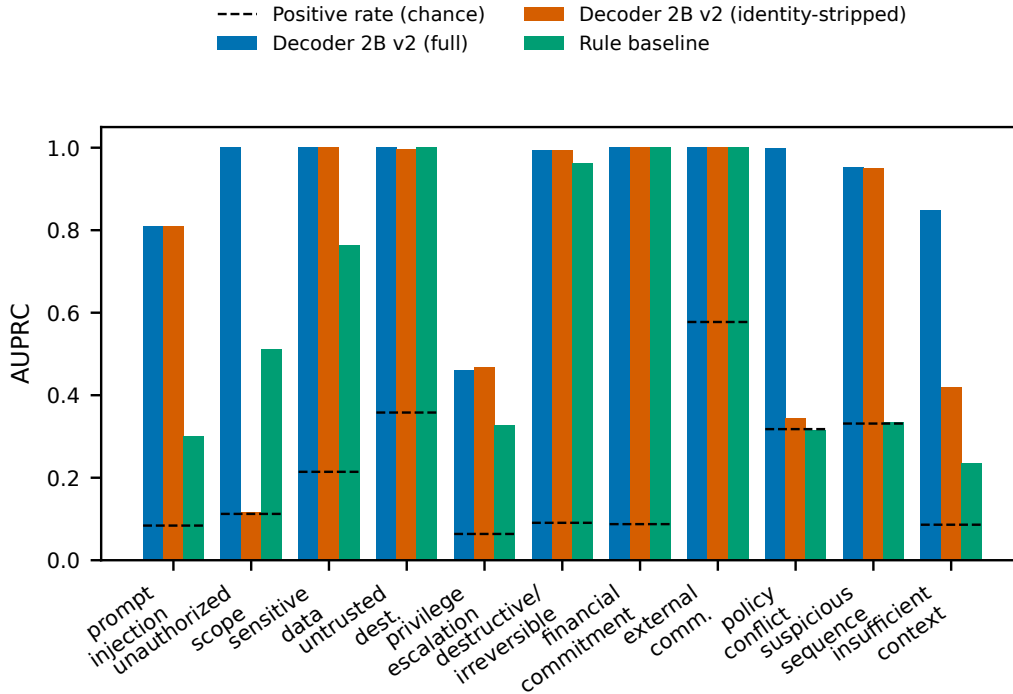


Figure 1: Decoder 2B v2, per-dimension AUPRC on `heldout_family` ($n=6640$, synthetic data): full model vs. identity-stripped vs. rule baseline, with each dimension’s positive rate (chance level) marked by a dashed line. Source: `docs/results/synthetic-v2/decoder-2b/`.

The macro drop is concentrated entirely in the three delegated-authority dimensions. `unauthorized_scope` falls below the rule baseline (0.116 vs. 0.512) once the entitlements it must be checked against are gone; `policy_conflict` falls to 0.345, barely above the 0.315 rule floor, because there is no clause left to violate; `insufficient_context` falls by half. The fourth row is untouched: its deciding fact renders directly into the action arguments regardless of identity, and the rule baseline already gets it right. Four more dimensions (`destructive_or_irreversible_action`, `untrusted_destination`, `external_communication`, `sensitive_data_exposure`) show the same flat, lookup pattern; two more (`prompt_injection_influence` 0.810, `suspicious_action_sequence` 0.952) are learned from trajectory order and are identity-independent.

The zero-shot models’ near-chance macro AUPRC (0.24–0.27), unmoved by identity, shows this is not a prompting gap: both were trained for harm, jailbreak, and groundedness criteria, not for checking an action against entitlements or policy. `gpt-oss-safeguard` is also $\sim 50\times$ slower per row than the 2B decoder, for no benefit on the dimensions that matter here.

4 POLICY GENERALIZATION: KIND VS. PHRASING

Robustness to unseen *tools* (`heldout_family`) is not robustness to unseen *policy kinds* or *phrasings*, and the latter is the more product-relevant test: a deployment needs the classifier to apply a customer’s own wording, not just handle a new tool under a training-set policy. ADR 0010 adds two splits: `heldout_policy_kind` withholds entire predicate kinds (e.g. `forbid_recipient_domain`) from training; `heldout_policy_phrasing` withholds one paraphrase index of every kind the model *did* train on. v4 additionally fixes a renderer defect that left `privilege_escalation` undecidable from text (§8).

Figure 2 makes the kind-vs-phrasing asymmetry visible: bars for test, `heldout_family`, and `heldout_policy_phrasing` sit close together near the ceiling, while `heldout_policy_kind` drops toward, though still above, the rule baseline line.

Two very different results. **Unseen phrasing generalizes.** A paraphrase index never seen in training leaves `policy_conflict` at 0.987–0.993 AUPRC, matching the in-distribution test split (0.994): the model reads the clause’s meaning, not a template id. **Unseen policy kinds do not.** With two of fifteen kinds withheld, `policy_conflict` drops to 0.653 (v3) / 0.752 (v4) at a 0.32–0.41 rule-baseline floor – above chance but far below every kind seen in training, including under novel wording. Every other dimension on this split is unaffected, so the model has learned the fifteen kinds it saw, not the general operation of checking a predicate against a rendered fact.

The v4 fix also confirms `privilege_escalation`’s earlier AUROC-high/AUPRC-low signature (§8) was a data defect, not a model limit: rendering the deciding `authority_before/authority_after` values lifts it from 0.505 to 1.000 AUPRC on `heldout_family`.

ADR 0011 tests whether more kinds fix kind-generalization: it triples the inventory to 30 and withholds only 4 (a smaller fraction than v3/v4’s 2-of-15). A rise toward the phrasing-generalization

Table 3: Decoder 2B, macro AUPRC and `policy_conflict` AUPRC across splits, on the v3 (ADR 0010, 15 policy kinds, 2 withheld) and v4 (same data, `privilege_escalation` render fix) generators. v5 (ADR 0011, 30 kinds, 4 withheld) tests whether more kinds let the model generalize the *shape* of policy-conflict reasoning rather than memorize kinds. Source: [docs/results/synthetic-v2/README.md](#).

Split	v3 macro	v3 <code>policy_conflict</code>	v4 macro	v4 <code>policy_conflict</code>
test	0.908	0.994	0.960	0.994
<code>heldout_family</code>	0.914	0.941	0.960	0.964
<code>heldout_policy_kind</code>	0.889	0.653	0.944	0.752
<code>heldout_policy_phrasing</code>	0.921	0.987	0.968	0.993
v5 (30 kinds, 4 withheld)	[TODO: pending: train-2b-b200-v5-recipe.yaml]			

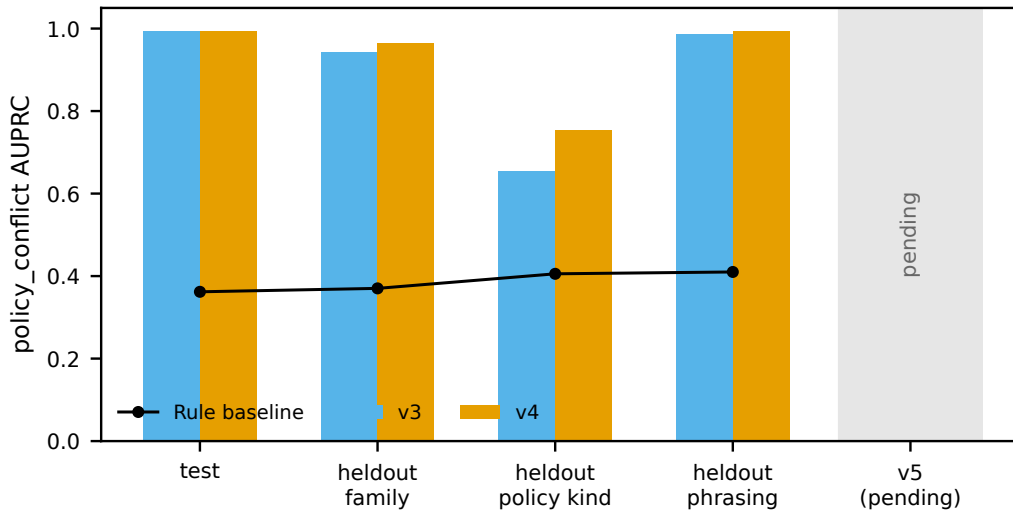


Figure 2: Decoder 2B, `policy_conflict` AUPRC across splits for the v3 and v4 generators (synthetic data; n per split matches Table 3), with the rule baseline’s `policy_conflict` AUPRC on each split shown as a line. The v5 slot is left blank pending that run. Source: docs/results/synthetic-v2/decoder-2b-v3/, decoder-2b-v4/.

level (~ 0.99) would make “kind” a learnable unit of generalization; staying in the 0.65–0.75 band despite more diversity would point to a structural limit needing retrieval-plus-entailment rather than more kinds. Pending at submission time (Table 3, last row).

5 MULTI-POLICY COMPOSITION AND THE CALIBRATION HINGE

Guardrail deployments commonly run several independently-tuned policies at once and act on the most restrictive decision – OR-ing k binary guards. Each guard’s false-positive rate compounds under that OR purely as a function of k , independent of any guard getting worse. forecheck’s alternative is one calibrated probability vector over the eleven dimensions, composed by one policy engine, decided either by per-dimension thresholds (`threshold` mode) or by the argmin of expected cost over the joint vector (`expected_cost` mode).

Figure 3 shows the independent and joint threshold strategies’ FPR climbing sharply as k grows past 2, while `expected_cost_joint` stays flat; the encoder’s `expected_cost_joint` curve, by contrast, sits near the top of the FPR panel throughout, illustrating that the flat curve is a property of the decoder’s calibration, not of the decision rule alone.

Threshold-based independent stacking pushes the decoder’s FPR from 0.015 at $k=1$ to 0.966 by $k=3$, staying there through $k=8$ – effectively denying everything, not because any policy got worse but purely because more were added. Deciding from the joint calibrated probability vector under

Table 4: Decoder 2B, false positive rate on covered-benign examples, `heldout_family` ($n=6640$), stacking k synthetic per-dimension policies. `independent` ORs k separately-thresholded decisions; `expected_cost_joint` merges the same k policies into one bundle and decides from the joint probability vector. Full table with the `joint` strategy, FNR, review rate, and the granite-embedding-r2 encoder is Table 6 in the appendix. Source: docs/evaluation/multi-policy-stacking.md.

k	1	2	3 (8 identical)
independent FPR	0.015	0.140	0.966
expected_cost_joint FPR	0.015	0.082	0.082

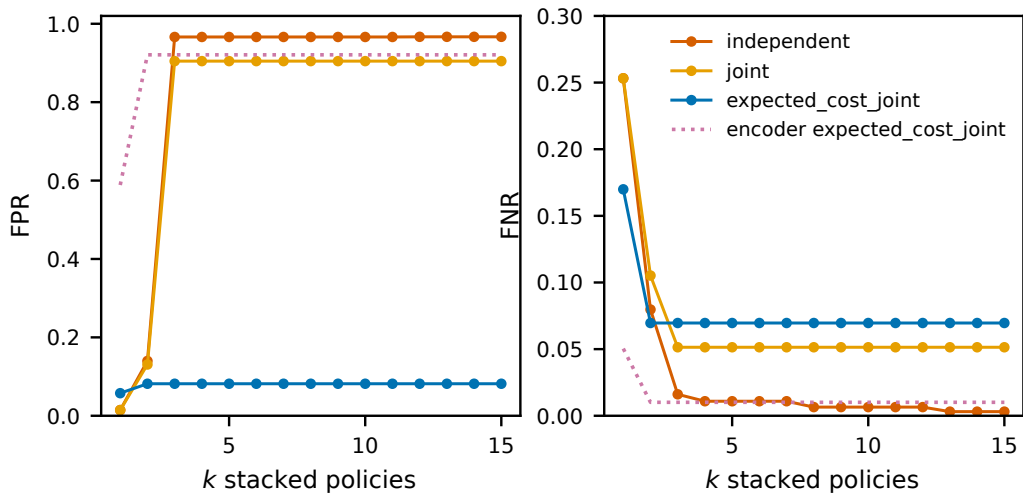


Figure 3: Decoder 2B, FPR (left) and FNR (right) vs. number of stacked policies k on `heldout_family` ($n=6640$, synthetic data), for independent, joint, and `expected_cost_joint` strategies; the granite-embedding-r2 encoder’s `expected_cost_joint` curve is shown dotted for comparison. Source: `docs/results/synthetic-v2/decoder-2b/heldout_family-stacking-report.md`, `encoder-granite-embedding-r2/heldout_family-stacking-report.md`.

`expected_cost` is invariant to k from $k=2$ on (FPR/FNR/review rate flat at 0.082/0.070/0.28): the decision depends on the probability vector and a fixed cost table, not on how many rules mention a dimension, so stacking more policies cannot by itself make the classifier more conservative.

The invariance is a calibration property, not a decision-rule property. The same rule on the granite-embedding-r2 encoder’s probabilities – identical examples and policies – settles at FPR 0.921, not 0.08 (Table 6). The decoder is well calibrated (macro ECE 0.004 on `heldout_family`, temperature-scaled on the calibration split (Guo et al., 2017; Platt, 1999)); the encoder is flatter (ECE 0.049). Expected-cost composition only keeps a stack from collapsing when the probabilities feeding it are sharp and calibrated, and cannot be recovered by rule engineering alone on a poorly-calibrated model.

The cost table used throughout is the default severity-tier table, not tuned to any deployment: the $k=1$ row already trades false positives for false negatives relative to the F1-optimal threshold (FPR/FNR 0.058/0.170 vs. 0.015/0.253). The k -invariance claim is about the *shape* of the composition curve, not that this default cost table is right for any given customer.

6 CAN THE ACTING AGENT JUDGE ITS OWN SCOPE?

The classifier above is an external checker: it never generates the action, only scores it. A cheaper alternative is to ask the acting agent itself, from inside its own system prompt, whether to proceed. We give the same untrained instruct models used as the decoder’s base (2B, 8B) a Fin-style system prompt carrying the identical entitlements, delegated scope, and policy text, followed by the pending tool call, and read a single ALLOW/STOP verdict off the first output token. That one scalar is scored against all eleven dimensions’ labels: tracking the external checker means an undifferentiated act/refuse decision happens to catch that risk; lagging badly means the dimension needs external, per-dimension structure the agent’s own judgment does not supply.

Figure 4 shows the pattern holds across all eleven dimensions, not just the five in Table 5: the agent’s bars track the chance line closely everywhere except `policy_conflict` and `external_communication`, while the external checker clears 0.84 AUPRC or better on every dimension.

Table 5: Agent self-judgment (8B, untrained, single ALLOW/STOP verdict) vs. the trained decoder external checker (v4), AUPRC on heldout_family ($n=2000$ for the agent rows). Positive rate for each dimension in parentheses. Source: docs/results/synthetic-v2/README.md.

Dimension (pos. rate)	Agent 8B	Agent 8B, stripped	Decoder 2B v4 (checker)
unauthorized_scope (0.106)	0.138	0.104	0.999
policy_conflict (0.385)	0.509	0.415	0.964
insufficient_context (0.079)	0.085	0.084	0.841
privilege_escalation (0.075)	0.103	0.106	1.000
prompt_injection_influence (0.075)	0.169	0.498	0.81 ^a
Macro	0.242	0.281	0.960

^aDecoder 2B v2 number, the version this comparison’s original report used; v4 not yet re-run on this dimension in the self-judgment comparison.

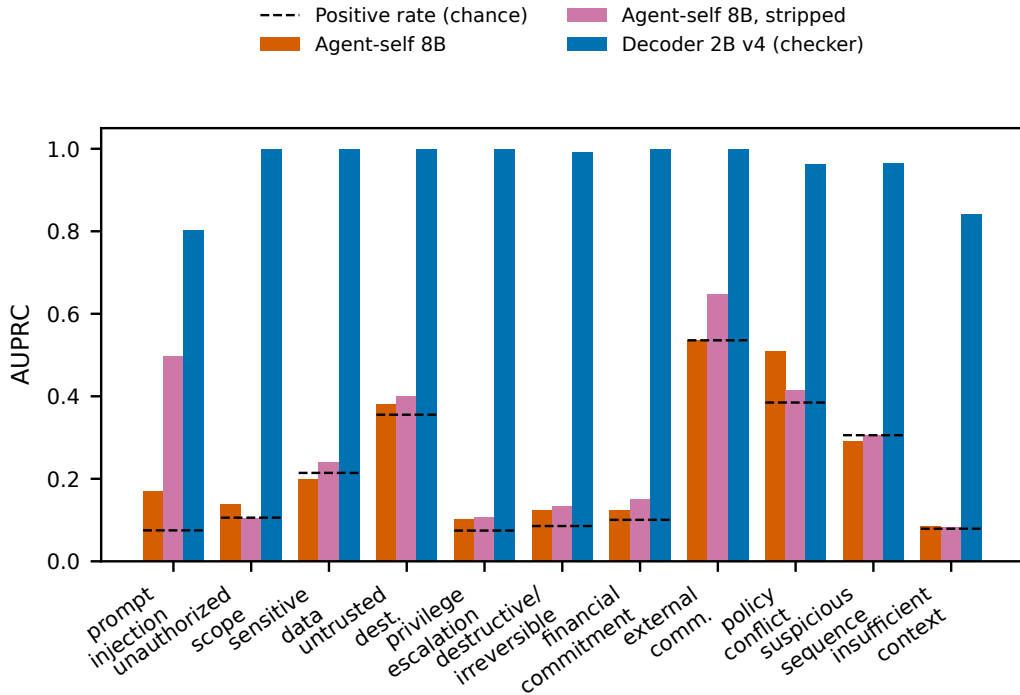


Figure 4: Per-dimension AUPRC on heldout_family (synthetic data; $n=2000$ for the agent-self-8B rows, $n=6653$ for the decoder), with each dimension’s positive rate (chance level) marked by a dashed line. Source: docs/results/synthetic-v2/agent-self-8b/, decoder-2b-v4/heldout_family-report.md.

Two findings. First, **the agent cannot judge its own scope from inside its prompt**: with full identity context the 8B agent is at or near chance on every identity-dependent dimension (AUROC 0.53–0.67, not shown), the same level as the zero-shot safety models in §3. The delegation information sits in its own prompt and is not turned into a decision. Second, **identity context makes the agent worse, not better, at spotting injection**: stripping the policy and scope text raises `prompt_injection_influence` AUPRC from 0.169 to 0.498 – a long policy preamble appears to dilute attention to an injected instruction elsewhere in the trajectory. Both argue for keeping an external checker in the loop rather than trusting the agent’s own judgment, even on content-style dimensions. The small identity signal that does exist (`unauthorized_scope` AUROC 0.61 full vs. 0.48 stripped) shows the agent reads the context a little; it does not turn that into a usable decision, and the gap to the external checker’s 0.960 macro AUPRC persists regardless.

7 RELATED WORK

LLM guardrails. Llama Guard (Inan et al., 2023), Granite Guardian (Padhi et al., 2024), ShieldGemma (Zeng et al., 2024), NVIDIA Aegis (Ghosh et al., 2024), and gpt-oss-safeguard (OpenAI, 2025) are general-purpose content/harm classifiers over a conversation turn or action. We do not claim they fail at their intended purpose; §3 shows they are near chance specifically on delegated-authority dimensions (`unauthorized_scope`, `policy_conflict`, `insufficient_context`) they were not trained to answer, regardless of whether identity context is in their prompt.

Agent safety benchmarks. AgentHarm (Andriushchenko et al., 2024), ToolEmu (Ruan et al., 2023), InjecAgent (Zhan et al., 2024), and AgentDojo (Debenedetti et al., 2024) measure end-to-end agent behavior or injection success via an outcome judge, not principal entitlements or delegated scope as structured input, and produce no per-action calibrated probability a downstream policy engine can compose (§2, §5): they answer "did the agent do something bad," not "should this action have been allowed given who it was for."

Authorization and policy engines. OPA (Open Policy Agent), Cedar (Cedar Policy), and XACML (OASIS, 2013) are deterministic, symbolic engines: given a fully specified request and policy, they decide allow/deny with no learned uncertainty. forecheck’s policy layer is architecturally in this family, but the composition problem in §5 exists only because its input is a learned, calibrated probability rather than a fact a symbolic engine evaluates exactly.

Calibration. Platt/temperature scaling (Platt, 1999), the modern re-examination of network calibration (Guo et al., 2017), and the Wilson score interval (Wilson, 1927) (used for the approval-elimination bound, Appendix A) are standard tools we apply directly to policy *composition*, not where this literature is usually pointed: §5’s k -invariance result is a direct function of the underlying model’s calibration sharpness.

8 LIMITATIONS

Synthetic data only. Every number here is on data from forecheck’s own scenario generator (`docs/evaluation-plan.md`, classes 1–2). No human-labelled or production-traffic evaluation exists at submission time, and no real-world safety claim is made: every result measures whether an arm learned the generator’s own label function.

Generator regularities. The model may be exploiting template regularities in how the generator writes policy clauses and entitlement lists, rather than a general policy-reading ability. The `heldout_policy_kind/heldout_policy_phrasing` splits (§4) probe this and only partly rule it out: phrasing generalizes cleanly, kind does not, consistent with a lookup over kinds rather than the underlying operation. The encoder’s earlier null result on both splits is uninformative the same way – it never learned `policy_conflict` at all, so a flat curve shows nothing.

privilege_escalation render defect (v2/v3). The `authority_before/authority_after` fields deciding this label were never rendered into the tool-call text in v2/v3, so every arm’s 0.46–0.53 AUPRC there reflects a data defect: 78 of 79 dev positives were textually indistinguishable from an in-scope re-grant. Fixed in v4 (§4); numbers sourced from v2/v3 elsewhere should be read with this in mind.

Single seed. Every trained-arm result is from one training seed. A seed-variance re-run [TODO: pending: decoder-2b-v4-seed1] is needed before these point estimates should be read as more precise than the seed-to-seed variance they have not yet been measured against.

Held-out-policy-kind failure. `policy_conflict` AUPRC drops from ~ 0.99 (in-distribution, unseen-phrasing) to 0.65–0.75 on entirely unseen kinds. The supportable claim today is "one model, policies of a kind seen in training, any phrasing" – not "arbitrary new policy kinds." The v5 experiment (ADR 0011) testing whether a larger kind inventory fixes this is pending.

Other caveats. A single generator family (English, template renderer) produces every example. The §5 cost table is the default severity tier, not tuned to any organization. Generator seed nondeterminism was found and partially fixed mid-project (docs/results/notes/privilege-escalation-diagnosis.md); v2/v3 data are not byte-reproducible from their configs.

9 CONCLUSION

Pre-execution risk for agent tool calls is mostly a delegated-authority problem, not a content-safety problem, and general-purpose safety models do not solve it: stripping identity and policy context from a trained classifier’s input collapses exactly the dimensions that depend on it, and two zero-shot safety models are at chance on those same dimensions whether or not the context is available to them. Identity/delegation context and calibration quality are not incidental engineering details for agent guardrails – they are the two properties that determine whether a guardrail can detect authorization violations at all (§3) and survive being deployed alongside more than one policy (§5). What remains open is whether policy-conditioned reasoning generalizes to genuinely novel policy kinds, not just novel wording of familiar ones (§4); the v5 experiment testing this, and a first real-data probe against hand-labelled traffic, are the next things this line of work needs before any claim above can be extended past synthetic data.

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A APPENDIX

A.1 FULL MULTI-POLICY STACKING TABLE

Table 6 extends Table 4 with the `joint` strategy (rules from all k policies merged into one bundle and evaluated once in threshold mode, rather than decided independently and OR-ed), false negative rate, review rate, and the granite-embedding-r2 encoder run on the identical examples and synthetic per-dimension policies. `independent` and `joint` are mathematically identical at every k here because the synthetic stacking policies use only `plain`, `non-hard`, `non-allow_override` rules, so there is nothing for the engine’s combination logic to do differently between the two strategies on this construction (`docs/evaluation/multi-policy-stacking.md` §3); the two rows diverge only once real bundles with `hard/allow_override` rules are mixed in, which is outside this diagnostic.

Note the encoder’s `expected_cost_joint` FPR (0.921) is *worse* than its own independent FPR at $k=2$ (0.710): on a poorly calibrated probability vector, aggregating across dimensions does not reliably help and can hurt, underscoring that the composition benefit in §5 is conditional on calibration quality, not a property of the `expected_cost` rule in isolation.

Table 6: Multi-policy stacking, heldout_family (n=6640). Source: docs/results/synthetic-v2/README.md, docs/evaluation/multi-policy-stacking.md.

Arm	k	Strategy	FPR	FNR	Review rate
Decoder 2B	1	any	0.015	0.253	0.31
Decoder 2B	2	independent	0.140	0.080	0.30
Decoder 2B	2	expected_cost_joint	0.082	0.070	0.28
Decoder 2B	3	independent	0.966	0.016	0.66
Decoder 2B	3	joint	0.905	0.051	0.63
Decoder 2B	3	expected_cost_joint	0.082	0.070	0.28
Decoder 2B	8	independent	0.966	0.007	0.58
Decoder 2B	8	expected_cost_joint	0.082	0.070	0.28
Encoder granite-r2	2	independent	0.710	0.042	0.26
Encoder granite-r2	2	expected_cost_joint	0.921	0.010	0.37
Encoder granite-r2	3	independent	0.990	0.008	0.21
Encoder granite-r2	3	expected_cost_joint	0.921	0.010	0.37

Table 7: Decoder 2B v2, approval elimination on heldout_family (n=6640), bundle balanced. “Reweighted” importance-reweights rows so the weighted incident rate equals 5%, matching a plausible deployment rate rather than this adversarial split’s own 58% base rate. v4 curves are [TODO: pending: decoder-2b-v4 -approval-curve] at submission time. Source: docs/results/synthetic-v2/README.md, docs/evaluation/approval-elimination.md.

Budget	Eliminated (unweighted, base 0.58)	Eliminated (reweighted to 5%)	Review rate (rw)
0.1%	0.000	0.000	0.975
0.5%	0.000	0.000	0.975
1%	0.000	0.888	0.088
2%	0.000	0.961	0.017
5%	0.000	0.992	0.001

A.2 APPROVAL ELIMINATION

The product-facing metric is not classifier accuracy but how many of a deployment’s human-approval reviews can be safely automated at a fixed, chosen incident rate: sort examples by the decoder’s expected cost of ALLOW, and ask what fraction of the review queue can be auto-decided while the fraction of auto-allowed actions that are actually risky stays under a budget. Because the evaluated heldout_family split is adversarial-heavy (58% base incident rate), Table 7 reports both the raw unweighted curve and a curve importance-reweighted to a stated 5% deployment incident rate, with elimination at each budget computed conservatively via a 95% Wilson score upper bound (Wilson, 1927) on the realized incident rate in the auto-allowed set.

Figure 5 shows the three curves nearly overlapping: v2 and v4 eliminate a comparable share of approvals at every budget on heldout_family, and v4’s test split tracks its own heldout_family curve closely, so the elimination curve is not sensitive to the v3/v4 generator fix or to family-held-out tools.

At a deployment incident rate of 5%, the calibrated v2 decoder auto-allows 88.8–99.2% of actions that would otherwise need human review while keeping the allowed-but-risky rate at or below the stated budget. The unweighted column is near zero not because the model is uninformative but because the split’s own top-ranked rows already contain risky examples by construction (an adversarial benchmark, not a deployment population); it is a ceiling of the split’s design, not of the classifier. Below a 1% budget the Wilson bound cannot be met at this split’s effective sample size and the conservative curve reports zero elimination; a point-estimate mode is available for that regime but is dominated by single rows at small allow-set sizes.

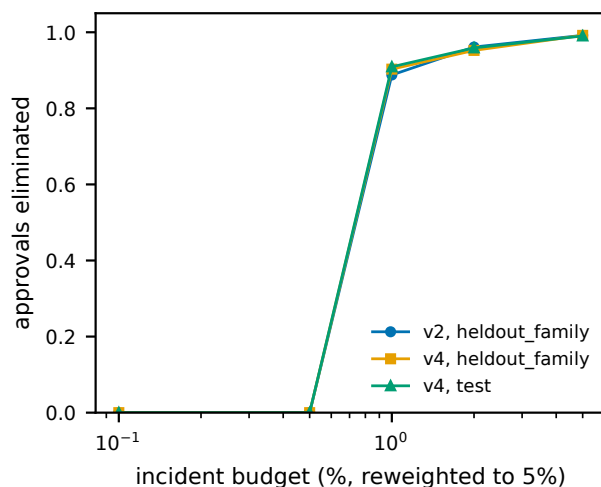


Figure 5: Approvals eliminated vs. incident budget, reweighted to a 5% deployment incident rate, for decoder 2B v2 (heldout_family, $n=6640$) and v4 (heldout_family $n=6653$, test $n=4412$), all synthetic data, log-scaled budget axis. Source: docs/results/synthetic-v2/decoder-2b/heldout_family-approval-report.md, decoder-2b-v4/{heldout_family,test}-approval-report.md.

A.3 EXPERIMENT MATRIX

The full list of experiments this paper still needs, with recipe files and status, is tracked in paper/EXPERIMENTS.md rather than duplicated here.